Measurement: Homework 2

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1 Child Mortality in Côte d'Ivoire (6 points)

1. Sampling (1 points)

(a) How should you transform pweight to obtain a weight that allows you to estimate statistics on the population of women in Cote d'Ivoire? Explain your computations.

I can use the pweight column to balance the class distribution of the sample to match the population distribution. The `pweight` column contains sample weights that modify the contribution of each observation to the overall statistics. It works as a frequency multiplier, where each observation is counted as if it were `pweight` number of observations. In practice, R can use the pweight column to compute weighted statistics, such as means and proportions by using the `weights` argument in functions like `lm()` or `survey::svydesign()`. Internally, weighted functions such as weighted means and lm will multiply each observation by its corresponding weight. For example a weighted mean will be computed using equation Equation 1, and weighted linear regression will minimize the weighted sum of squared residuals, as shown in equation Equation 2.

$$\bar{x}_w = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i} \tag{1}$$

$$\sum_{i=1}^{n} w_i (y_i - \hat{y}_i)^2 \tag{2}$$

2. Measurement of under 5 mortality rate (5 points)

(a) What method can be deemed preferable and why? Give the pros and cons of both methods.

The preferable method is method 2 (M2) because method 1 (M1) underestimates the mortality rate by not capturing the deaths that occur after the survey period. Figure 1 shows graphycally the surveyed groups: the x-axis represents the age of the child at the time of the survey, and the y-axis represents the age of the child at the time of death. The red arrow at 45 degrees indicates the frontier of possible outcomes, since by definition noone can die after their current age, the survey caputures outcomes below the frontier. Notice M1 captures area B and drops area C underestimating the true mortality rate. M2 captures area C which represents the true mortality rate. For example a surveyed mother has a child one year before the survey and the child dies the year after the survey (area C in Figure 1), M1 will not capture his death because by definition nothing can die after its current age.

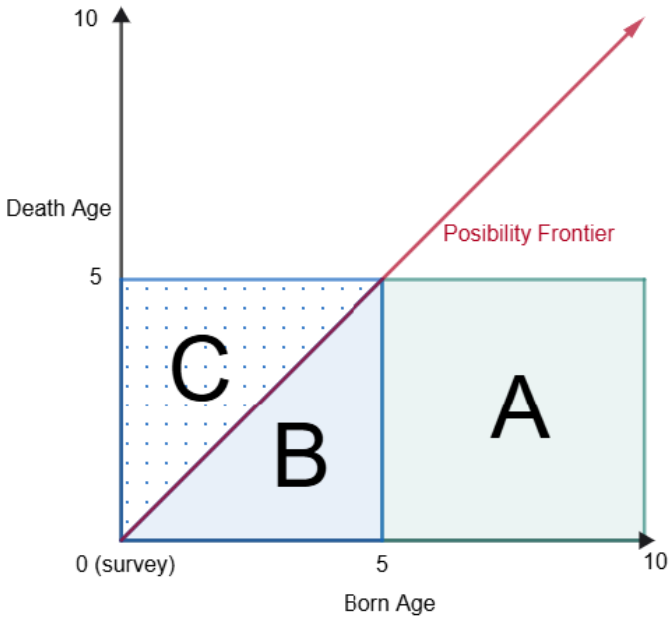
M1

- Pros: it is easy to compute, captures the most recent deaths (recency).
- Cons: underestimates the mortality rate, downward biased, does not capture deaths after the survey period.

M2

- Pros: provides a robust estimate of the true mortality rate.
- Cons: it requires a longer period of time to capture deaths, less recent than M1.

Figure 1: Mortality estimator analysis



(b) Child mortality rates in Côte d'Ivoire.

i. Compute mortality rates with both methods for boys and girls separately (don't forget the weights).

Table 1: Child mortality rates in Côte d'Ivoire, by sex and method.

	M1 (Boys)	M2 (Boys)	Diff (Boys)	M1 (Girls)	M2 (Girls)	Diff (Girls)	Diff M2 G-B
Mortality rate	0.117 ***	0.135 ***	0.017	0.084 ***	0.115 ***	0.031 ***	-0.020 *
	(0.007)	(0.008)	(0.011)	(0.007)	(0.008)	(0.010)	(0.011)
N	1431	1332		1428	1291		
R2	0.000	0.000		0.000	0.000		

Significance: *** p < 0.01; ** p < 0.05; * p < 0.1

ii. Compare and comment.

Table 1 shows the estimated child mortality rates in Côte d'Ivoire for boys and girls using both methods and their differences. The results confirm our theory of M1 underestimating the true mortality rate. We can clearly see the method estimates differences are positive indicating that M1 systematically underestimates the true mortality rate estimated by M2. Using the best method (M2), the average mortality rate for boys is 13.5% and for girls is 11.9%. The difference between girls and boys is -2 percentage points and statitically significant at the 90% confidence level, indicating that boys have a higher mortality rate than girls.

2 Health status, nutritional status and wealth in Côte d'Ivoire (7 points)

Stunted children are defined as children showing a height stature that is lower than the median of the reference population by 2 standard deviations measured on the same reference population. That is children such as the z-score is lower than -2 ($z \leq -2$), where the z-score is given by the following formula:

$$Z = \frac{\text{height} - \text{WHOMedian}(\text{sex}, \text{age})}{\text{WHOstdev}(\text{sex}, \text{age})}$$

1. Weight Unit Correction

In 2008, the weight could be measured in either kilograms or hectograms. However, the unit was not systematically reported. It was therefore assumed that all the weights were reported in kilograms.

(a) Derive an expression for the sample mean weight in this context, and show that, if we knew just one parameter, we could obtain a non-biased estimate of the mean weight despite the hectogram issue. (.5 point)

Let's denote the estimated mean weight as W_{obs}, the true mean weight as W_{true}, and the proportion of weights measured in hectograms as p_h. The observed mean weight can be expressed as the sum of the mean weights measured in kg and in hectograms multiplied by their corresponding probabilities:

$$\begin{aligned} E[W_{\text{obs}}] &= (1 - p_h) \cdot W_{\text{true}} + p_h \cdot W_{\text{true}} \cdot 10 \\ &= W_{\text{true}} \cdot (1 - p_h + 10p_h) \\ &= W_{\text{true}} \cdot (1 + 9p_h) \end{aligned}$$

Thus, having an estimate of p_h would allow us to correct the observed mean weight. Notice that if $p_h = 0$, then $E[W_{\text{obs}}] = W_{\text{true}}$, and if $p_h = 1$, then $E[W_{\text{obs}}] = 10W_{\text{true}}$.

For applying the correction, it is necessary to have a good estimate of p_h. In this case, it can be approximated by the proportion of weights above 25kg in 2008 which is 30.4% of the observations. Using this, we can correct the mean weight estimate by dividing the observed mean weight by $(1+9p_h)$. See 2 where the 2008 mean weight is corrected. After correction, the mean weight in 2008 is 11.48kg, which is consistent with previous years.

Table 2: Summary of Weight Estimates With Hectogram Correction

Year	Obs	Mean Weight	Mean Weight Corrected
1988	6475	12.46	12.46
1993	7109	11.94	11.94
2008	7333	42.87	11.48

(b) Come up with an empirical criterion to identify hectogram mistakes, and a graphical way to show the 2008 weight distribution, before and after correcting for mistakes, compared with the 1988 and 1993 weight distributions. (1 point)

By looking at the original distribution the weight measurements in 2008 show a wider distribution than the 1988 and 1993 distributions. The 2008 distribution shows a principal component at around 12kg but also shows multiple peaks at weight $>$ 25kg while the 1988 and 1993 distributions show a principal component at around 12kg with minimal outliers e.g. a few values below 1kg. See Figure 2 LHS.

After updating the 2008 weight distribution by dividing all weights $>$ 25kg by 10, the distribution shows no peaks after 25kg. See Figure 2 RHS.

2. Z-Score Computation

Using the variables medheight and stdheight, compute each child's z-score. (.5 point) Estimated z-scores are computed using the mentioned formula. The resulting distribution of z-scores for 2-4 years old children by year and region is shown in Figure 3a.

Figure 2: Distribution of Weight (kg) Before and After Correction

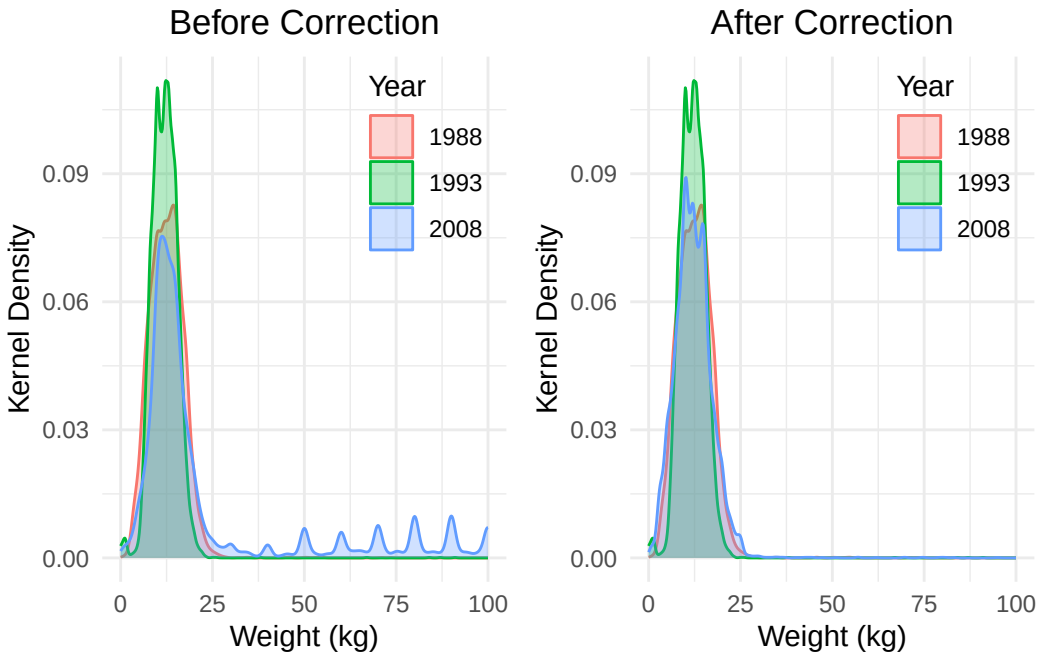
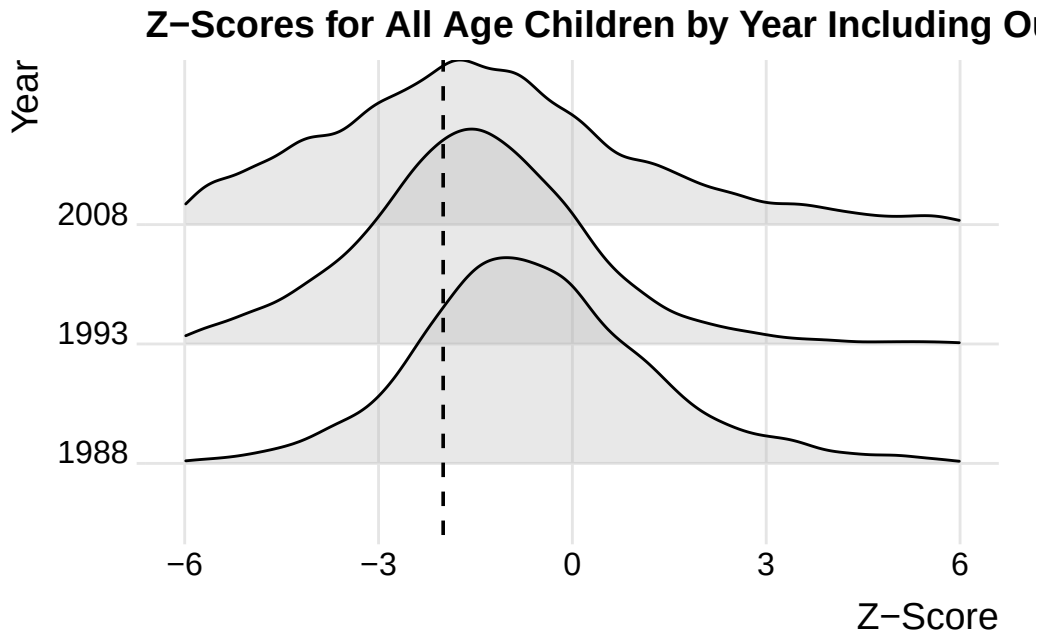


Figure 3: Distribution of z-scores by year

(a) X-axis is limited to -6 to 6, and the dashed line indicates the stunting threshold of -2.



3. Height Stature over time and across regions (3 points)

(a) Estimate the stunting rate for 2-4 years old children over time in the North, in the South, and at the national level. Decide what are outliers, and show results both including and excluding outliers. Comment your results, making all relevant comparisons.

As shown in Table 3, the national stunting rate for 2-4 years old children with outliers is 26.3% in 1988, 43.9% in 1993, and 47.1% in 2008. This increasing trend shows an overall increment of 20.8 percentage points in the stunting rate over the 20-year period.

The north and south regions an increasing trend overtime, but the north region shows a higher stunting rate than the south region in 1988 and 2008. Particularly, in 1988 the difference between north and south is 6 percentage points, and in 2008 the difference is 6.6 percentage points. In 1993 the difference is not statistically significant different frm zero, indicating that the north and south regions had similar stunting rates in 1993. Many theories can explain the differences between north and south e.g. differences in measurement, differences in the population, or the effect of socioeconomic factors such as poverty-alleviating programs, and access to health care.

The results excluding outliers show a similar trend in 1988 and 1993, however, in 2008 the stuting is lower in national level and in the south than in 1993. This change in the pattern suggest that the measurement correction done in question 1.b had an impact on the results and drafted conclusions. Regional differences between north and south persist, there are still significant differences of 6.6 and 6.2 percentage points in 1988 and 2008 between the north and south regions despite the change in the stunting rate over time pattern.

Table 3: National, North, and South stunting rates for 2-4 years old children excluding outliers.

	National	North	South	Diff (North-South)	National	North	South	Diff (North-South)
Year 1988	0.241 ***	0.282 ***	0.226 ***	0.056 ***	0.239 ***	0.284 ***	0.222 ***	0.061 ***
	(0.009)	(0.016)	(0.011)	(0.020)	(0.009)	(0.016)	(0.010)	(0.020)
Year 1993	0.420 ***	0.409 ***	0.423 ***	-0.014	0.412 ***	0.397 ***	0.416 ***	-0.020
	(0.007)	(0.014)	(0.008)	(0.017)	(0.007)	(0.014)	(0.008)	(0.017)
Year 2008	0.464 ***	0.498 ***	0.453 ***	0.045 **	0.410 ***	0.436 ***	0.402 ***	0.035 *
	(0.008)	(0.015)	(0.009)	(0.018)	(0.008)	(0.016)	(0.010)	(0.019)
Outliers	Yes	Yes	Yes	Yes	No	No	No	No
N	11755	3194	8561		10897	2930	7967	
R2	0.410	0.421	0.407		0.381	0.384	0.381	

Significance: *** p < 0.01; ** p < 0.05; * p < 0.1

(b) Estimate the stunting rate for 2-4 years old children over time for boys and for girls at the national level. Decide what are outliers, and show results both including and excluding outliers. Comment your results, making all relevant comparisons.

As shown in Table 4, the stunting rate for boys and girls is increasing over time for the model with outliers (LHS four models). Starting from 1988, boys and girls had a stunting rate of 27% and 25% with no significant difference. In 1993, boys and girls had a stunting rate of 46% and 41% with a significant difference of 5 percentage points indicating boys had a higher stunting rate. In 2008, boys and girls had a stunting rate of 48% and 46% respectively with no significant difference.

The stunting rates pattern without outliers (RHS four columns) flips in 2008. In this period, removing the outliers lowered the stunting rates from 47% with outliers to 40% without outliers. Differences between boys and girls differences are significant in 1993 only while there is no significant difference in 1988 and 2008.

Figure 4: Distribution of z-scores for 2-4 years old children by year and region

(a) X-axis is limited to -6 to 6, and the dashed line indicates the stunting threshold of -2.

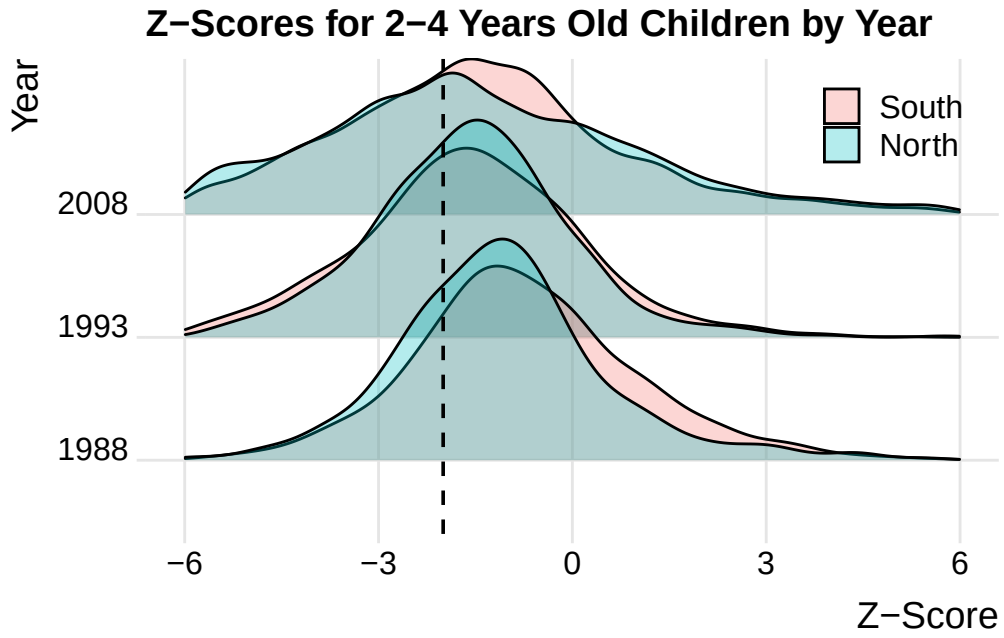


Figure 5: Distribution of z-scores for 2-4 years old children by year and region

(a) X-axis is limited to -6 to 6, and the dashed line indicates the stunting threshold of -2.

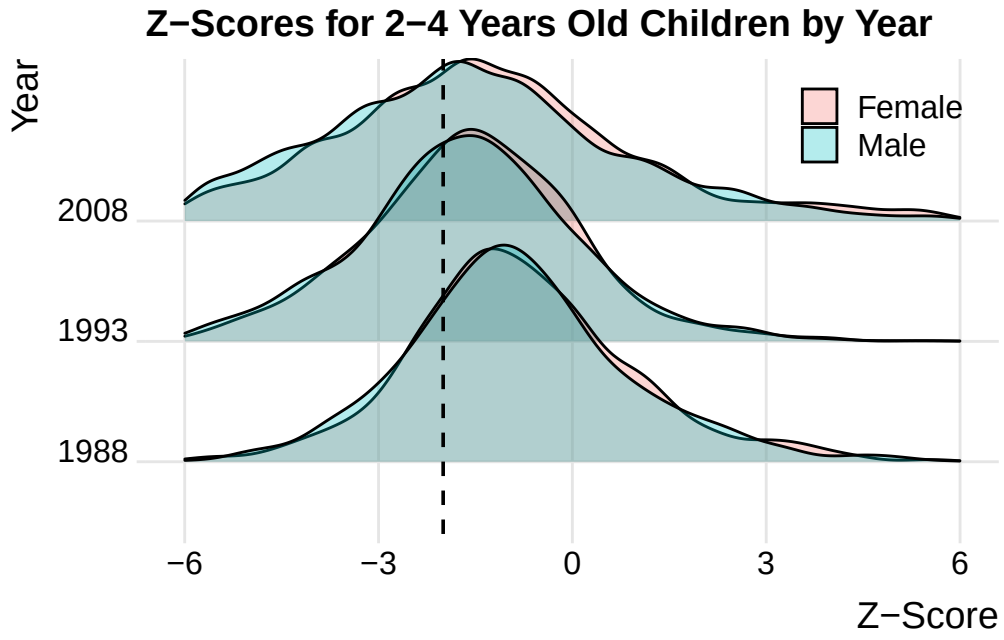


Table 4: National, Male, and Female stunting rates for 2-4 years old children.

	National	Male	Female	Diff (Male-Female)	National	Male	Female	Diff (Male-Female)
Year 1988	0.241 ***	0.246 ***	0.236 ***	0.011	0.239 ***	0.245 ***	0.233 ***	0.011
	(0.009)	(0.012)	(0.013)	(0.018)	(0.009)	(0.012)	(0.013)	(0.018)
Year 1993	0.420 ***	0.434 ***	0.405 ***	0.029 **	0.412 ***	0.425 ***	0.398 ***	0.028 **
	(0.007)	(0.010)	(0.010)	(0.014)	(0.007)	(0.010)	(0.010)	(0.014)
Year 2008	0.464 ***	0.473 ***	0.453 ***	0.020	0.410 ***	0.420 ***	0.398 ***	0.022
	(0.008)	(0.010)	(0.011)	(0.015)	(0.008)	(0.011)	(0.012)	(0.016)
Outliers	Yes	Yes	Yes	Yes	No	No	No	No
N	11755	6072	5683		10897	5610	5287	
R2	0.410	0.421	0.398		0.381	0.392	0.369	

Significance: *** p < 0.01; ** p < 0.05; * p < 0.1

4. Height Stature and poverty (2 points)

Here, one would like to study how the gradient of the number of stunted children aged 2-4 years, and of household consumption per capita, as well as the relationship between the two, have changed over time. Find a graphical or regression framework to represent this (many options are possible). Comment on your results and their limits.

[Your answer here]

3 A meaningful table (6 pts)

Construct a small statistical table showing a meaningful and policy relevant social fact about migration in the world. You may focus, or not, on a given region or country of origin, or on a particular subgroup of the population (children, etc.) A combination of originality, policy relevance and statistical accuracy will be very much appreciated. The table should fit in the top half of a page. It should have the attributes of a proper research-paper table: title, notes, etc. Table notes should indicate the source of data, the coverage (countries and population), the definition and units of variables and any other kind of information that is needed to read and understand the table. The bottom half of the page should be dedicated to a brief text, commenting upon the social fact that the table reveals.[cite: 31]

[Your table and text here]

Appendix: R Script

Please do not show your R script before every question, but put your entire R script at the end of the homework.